

Probabilistic Language Modeling Evaluation

Dr. Saeedeh Momtazi



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



آکادمی‌ها



- **Entropy**
- Entropy and Linguistics
- Language Model Evaluation
- Parameter Tuning and Cross-validation



- Entropy measures the amount of information in a RV
- Amount of information contained in a message (after removing all possible redundancy)
- number of bits that the message has after compression



$$H(V) = E[-\log(p(V))]$$

$$H(V) = \sum_{w_i \in V} -p(w_i) \log(p(w_i))$$

► Note: if you want the "unit" of the entropy to be "bit", you have to use the log to the basis2



Example 1

➤ Reporting the result of rolling an 8-sided die

➤ Entropy:

$$H(X) = - \sum_{i=1}^8 p(i) \log(p(i))$$

$$H(X) = - \sum_{i=1}^8 \frac{1}{8} \log\left(\frac{1}{8}\right) = -\log\left(\frac{1}{8}\right) = \log(8) = 3 \text{ bit}$$

The average length of the message needed to transmit an outcome of that variable using the optimal code



Example 1

- Reporting the result of rolling an 8-sided die
- The most efficient way is to simply encode the result as a 3 digit binary message:

1	2	3	4	5	6	7	8
001	010	011	100	101	110	111	000



Example 2

► Vocabulary with two words:

$V = a; b$

$$p(a) = x$$

$$p(b) = 1 - x$$

$$H = -x \log x - (1 - x) \log(1 - x)$$

$$x = 0 \rightarrow H = 0$$

$$x = 1 \rightarrow H = 0$$

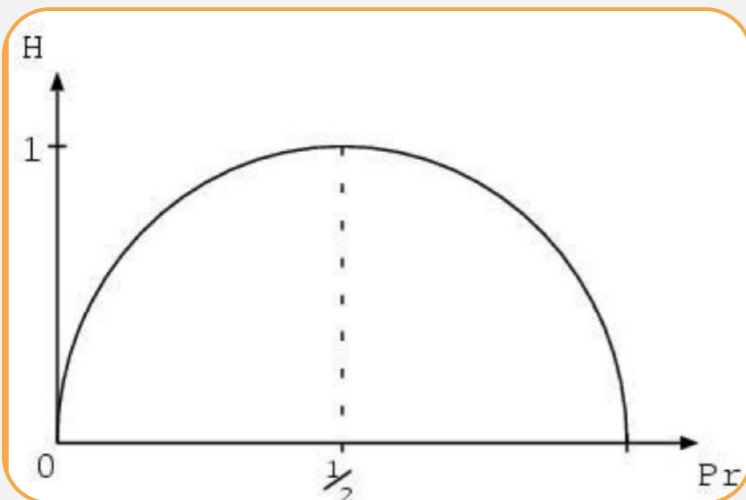


Example 2

► Vocabulary with two words:

$V = a; b$

$$H = -x \log x - (1 - x) \log(1 - x)$$





Example 3

➤ Vocabulary of W words w_i with uniform distribution $p(w_i) = \frac{1}{W}$

$$H = \sum_{i=1}^W -p(w_i) \log(p(w_i)) = \sum_{i=1}^W -\frac{1}{W} \log\left(\frac{1}{W}\right)$$

$$H = -W \frac{1}{W} \log\left(\frac{1}{W}\right) = -\log\left(\frac{1}{W}\right) = \log(W)$$

➤ Entropy for uniform distribution: log of the number of symbols



Joint Entropy

► The joint entropy of 2 RV X; Y is the amount of the information needed on average to specify both their values:

$$H(x) = - \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(x, y))$$



Conditional Entropy

- The conditional entropy of a RV Y given another X , expresses how much extra information one still needs to supply on average to communicate Y given that the other party knows X

$$H(Y|X) = \sum_{x \in X} p(x) H(Y|X = x)$$

$$H(Y|X) = - \sum_{x \in X} p(x) \sum_{y \in Y} p(y|x) \log(p(y|x))$$

$$H(Y|X) = - \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(y|x)) = -E(\log(p(Y|X)))$$



Chain Rule

$$H(X|Y) = H(X) + H(Y|X)$$

$$H(X_1, \dots, X_n) = H(X_1) + H(X_2|X_1) + \dots + H(X_n|X_1, \dots, X_{n-1})$$



Mutual Information

➤ $I(X, Y)$ is the mutual information between X and Y .

The reduction of uncertainty of one RV due to knowing about the other, or the amount of information one RV contains about the other

$$H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$$

$$H(X) - H(X|Y) = H(Y) - H(Y|X) = I(X, Y)$$



Mutual Information

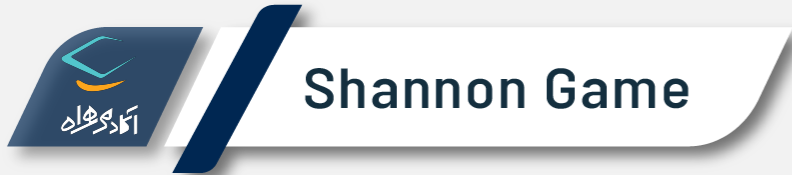
$$I(X, Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

► $I(X, Y)$ is 0 only when X and Y are independent:

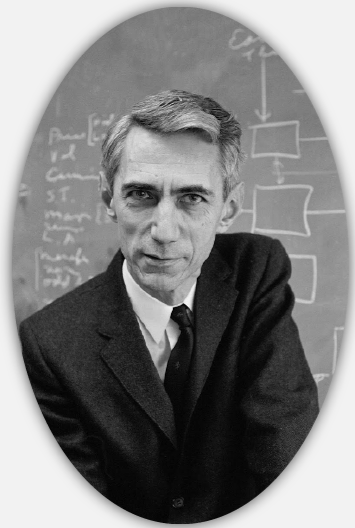
$$H(X|Y) = H(X)$$



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- **Entropy and Linguistics**
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► Shannon's Experiment to Calculate the Entropy of English



Claude Elwood Shannon

1916-2001

The Father Of Information Theory

<http://www.math.ucsd.edu/~crypto/java/ENTROPY/>



Complete the Sentence

► Th-r- -s -nly -n- w-y t- f-ll -n th- v-w-ls -n th-s s-nt-nc



Complete the Sentence

➤ Th-r- -s -nly -n- w-y t- f-ll -n th- v-w-ls -n th-s s-nt-nc

➤ There is only one way to fill in the vowels in this sentence



Entropy of a Language: Shannons Approach

- Show somebody the beginning of a text
- Ask him/her to guess the next letter
- Count the number of trials



Entropy and Linguistics

- Entropy is measure of uncertainty. The more we know about something the lower the entropy
- If a language model captures more of the structure of the language, then the entropy should be lower
- We can use entropy as a measure of the quality of our models



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- Definition:
- Perplexity is a measurement of how well a probability distribution or probability model predicts a sample.
- The perplexity of a discrete probability distribution p is defined as

$$2^{H(p)} = 2^{-\sum_{w_i \in V} p(w_i) \log(p(w_i))}$$



- In natural language processing, perplexity is a way of evaluating language models.
- A language model is a probability distribution over entire sentences or texts



Branching Factor

- Branching factor is the number of possible words that can be used in each position of a text
 - ▶ Maximum branching factor for each language is V



Branching Factor

➤ Branching factor is the number of possible words that can be used in each position of a text

▶ Maximum branching factor for each language is V

John eats an ...

apple

book

umbrella

orange

banana

desk

computer



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▶ Maximum branching factor for each language is V

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Branching Factor

- A good language model should be able to
 - ▶ minimize this number
 - ▶ give a higher probability to the words that occur in real texts



Branching Factor

Can we give the same knowledge
to a computer to predict the next character?



Perplexity

$$P(S) = P(w_1, w_2, \dots, w_n)$$

$$\text{Perplexity}(S) = P(w_1, w_2, \dots, w_n)^{-\frac{1}{N}} = \sqrt[N]{\frac{1}{P(w_1, w_2, \dots, w_n)}}$$

$$\text{Perplexity}(S) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_1, w_2, \dots, w_{i-1})}}$$

Goal: giving higher probability to frequent texts

→ **minimizing the perplexity of the frequent texts**



➤ The evaluation must give an indication of how well the learner will do when it is asked to make new predictions for data it has not already seen.

▶ Dividing the corpus into two parts

train

test

▶ Building a language model from the training set

▶ Estimating the probability of the test set

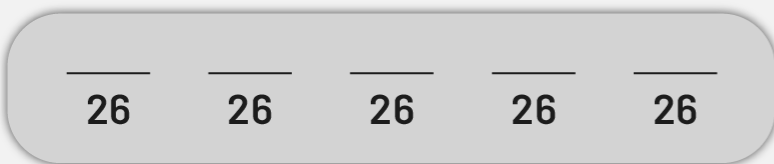
▶ Calculate the perplexity of the test set



➤ Maximum branching factor for each language is $|V|$

$$\text{Perplexity}(S) = \left(\prod_{i=1}^N P(w_i | w_1, w_2, \dots, w_{i-1}) \right)^{-\frac{1}{N}}$$

Example: predicting next characters instead of next words ($|V| = 26$)



$$\text{Perplexity}(S) = \left((1/26)^5 \right)^{-\left(\frac{1}{5}\right)} = 26$$



➤ Wall Street Journal

▶ Training set: 38 million word tokens

▶ Test set: 1.5 million words

	Unigram	Bigram	Trigram
Perplexity	962	170	109



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Normal Evaluation

- Dividing the corpus into two parts

train

test

- Building a language model from the training set
- Estimating the probability of the test set
- Calculating the perplexity of the test set



Evaluation with Parameter Tuning

- Dividing the corpus into three parts

train

dev test

- Building a language model from the training set
- Calculating the perplexity of the development set with different parameter values
- Choosing the best parameter value and use it to estimate the probability of the test set
- Calculating the perplexity of the test set



➤ There is no guarantee that the chosen test set is representative enough to model our data

➤ Solution:

- ▶ Assessing how the results of a statistical analysis will generalize to an independent data set
- ▶ Performing multiple rounds of cross-validation using different partitions, and the validation results are averaged over the rounds



Cross-validation

train

test

train

dev test

➤ k-fold cross-validation

➤ Leave-one-out cross-validation



Further Reading

➤ Speech and Language Processing (3rd ed. draft)

▶ Chapter 3